BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 207 · Complexity and algorithm choice

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Why is a queue suitable for a print server?

Q2. What condition must the array satisfy?

Q3. How many times does a for-loop from i=0 to i<6 execute?

Q4. A loop of length 4 inside another loop of length 4: how many inner operations?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Big O — Asymptotic growth of algorithm work.

O(1) — Constant growth.

O(log n) — Logarithmic growth, e.g. halving search intervals.

O(n) — Linear growth.

O(n^2) — Quadratic growth, e.g. complete nested scans.

New learning

Start with n — n is the number of items. More items usually mean more operations.

Trace loop nesting — One full scan performs n steps. A nested n-by-n scan performs n^2 steps.

Compare growth — When n doubles, linear work doubles; quadratic work roughly quadruples. Big O describes growth, not seconds.

Worked example

Compare operation counts of one loop and two nested loops for n=5.

1. Predict — One loop visits five elements, so does five operations.

2. Explain — The nested loop repeats five operations five times, making 25.

We Do · Practise with guidance

Compare linear and binary search for 64 ordered values.

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Linear: scan candidates.
Binary: discard half each step.
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Predict

Linear search can check all 64 values.

Explain

A standard midpoint binary search needs at most seven comparisons on 64 sorted values.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Give the Big O of a single full scan of n elements.

Question 2

For n=100, give iterations of a linear n loop and nested n×n loop.

Question 3

Why may linear search be preferable for a single search of a short unsorted array?

Exit Ticket · Answer before checking

$O(n^2)$ means an algorithm takes n^2 seconds.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. It processes jobs in arrival order (FIFO).

Q2. It must be sorted.

Q3. Six times.

Q4. 16.

You Do pages · Independent questions

Q1. O(n).

Q2. 100 and 10,000 respectively.

Q3. It does not need the sorting preparation required before binary search.

Exit Ticket · Model answer

False. It describes growth of work, not a fixed duration.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: